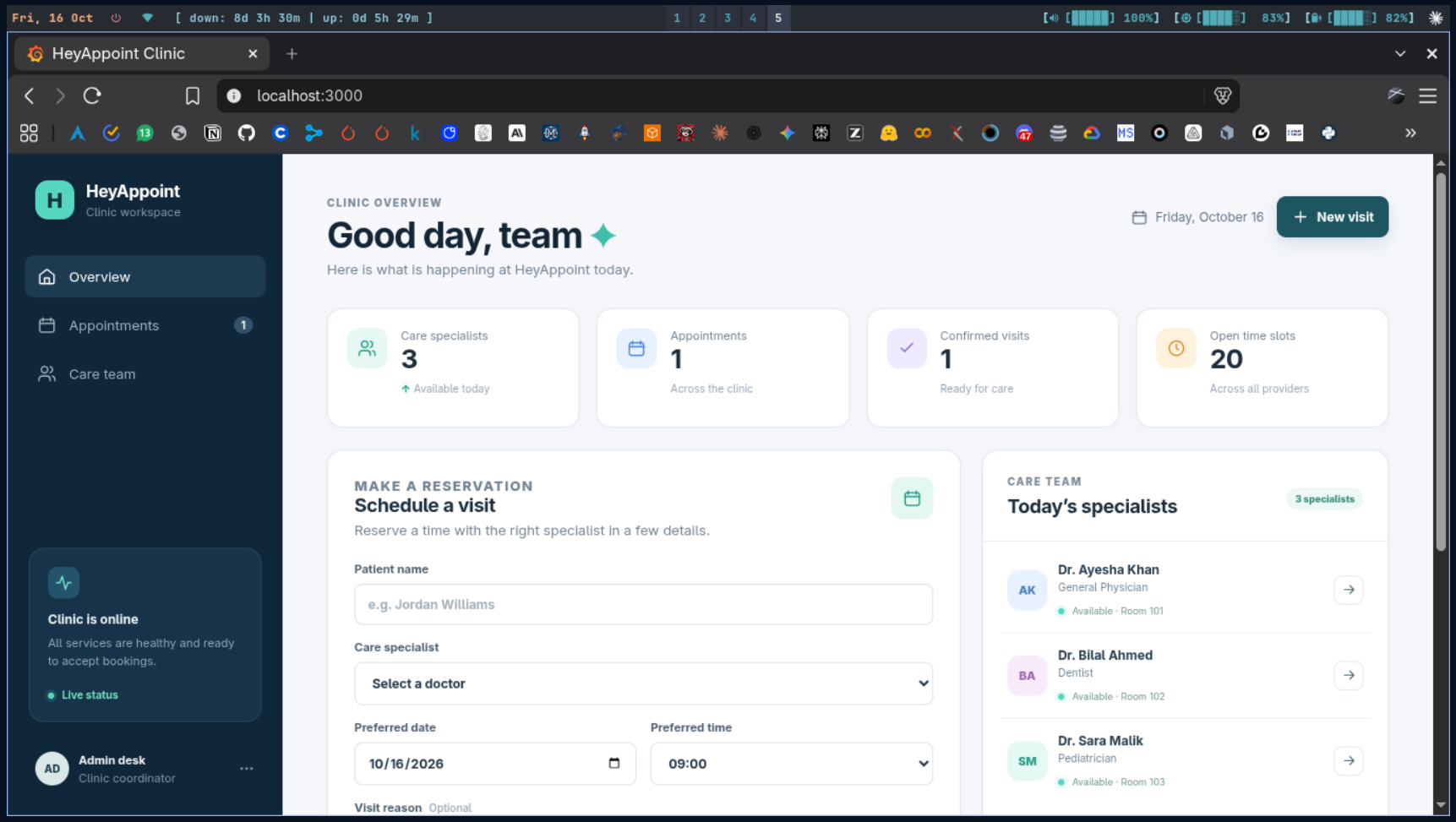
HEYAPPOINT CLINIC

A complete clinic appointment platform with a verifiable DevOps delivery system

Book. Operate. Observe. Ship safely.

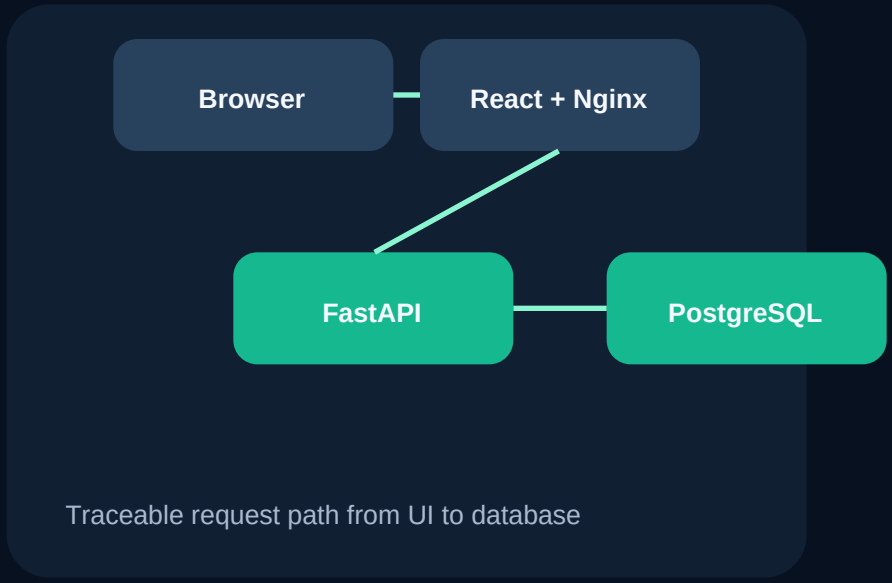
FastAPI + PostgreSQL + React + Docker + Kubernetes



THE PRODUCT

A simple workflow backed by real API and database behavior

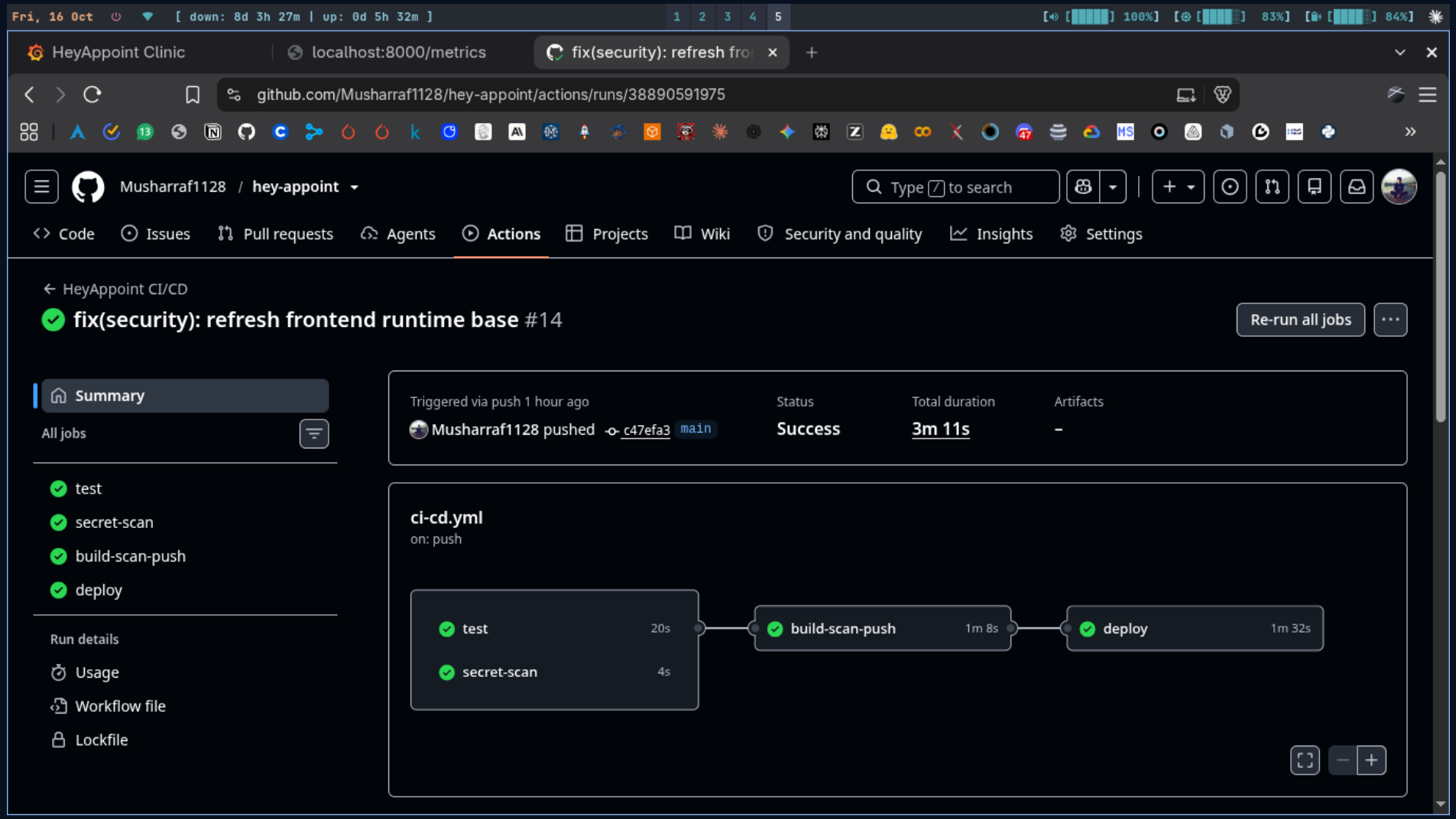
- Patients choose a doctor and time slot
- Staff confirm or cancel appointments
- Duplicate slots return HTTP 409
- GET, POST, PUT, and DELETE endpoints
- Alembic manages PostgreSQL schema changes



Traceable request path from UI to database

QUALITY AND DELIVERY GATES

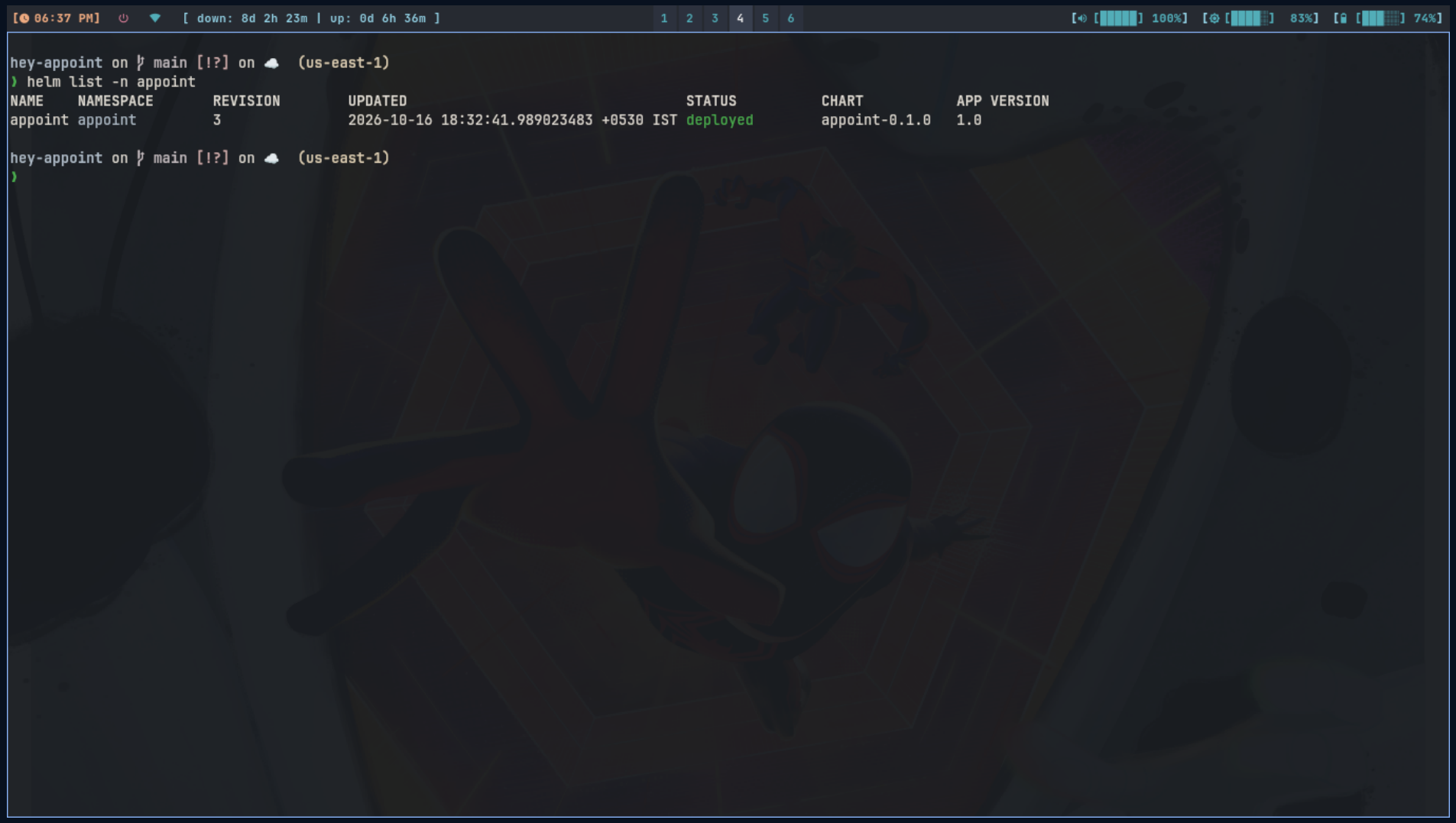
Automated checks run before an image reaches GHCR



TEST → BUILD → TRIVY → SHA-TAGGED GHCR → HELM SMOKE DEPLOYMENT

KUBERNETES AND HELM

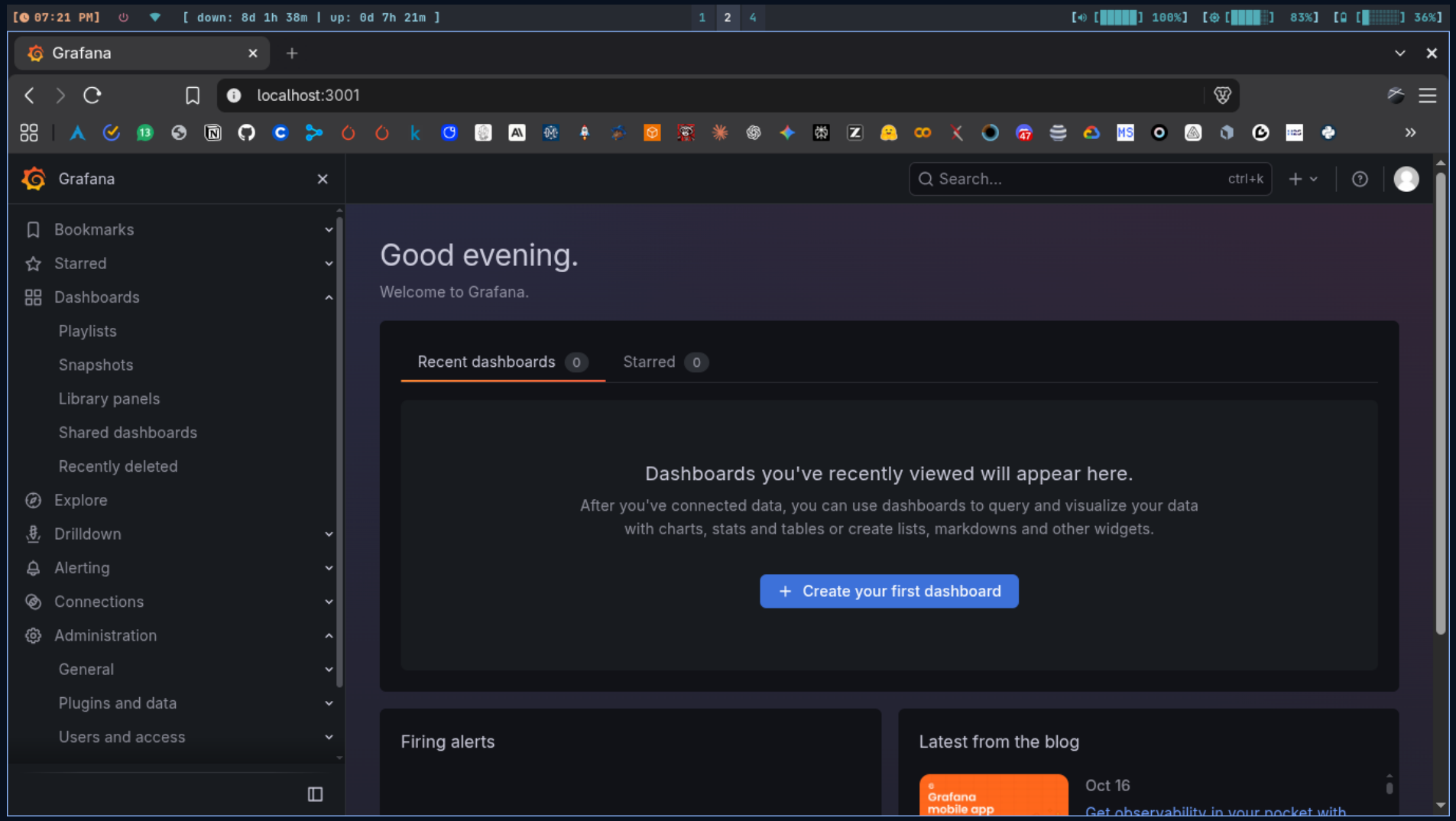
Reproducible deployment with health checks and service routing



Two application replicas • ClusterIP services • /api Ingress route • HPA • readiness/liveness probes

OBSERVABILITY

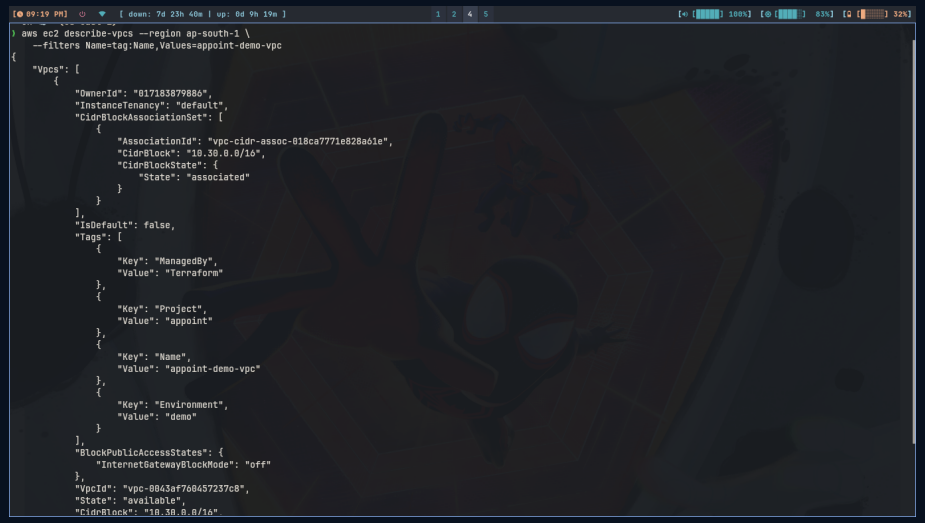
Prometheus and Grafana turn application behavior into visible signals



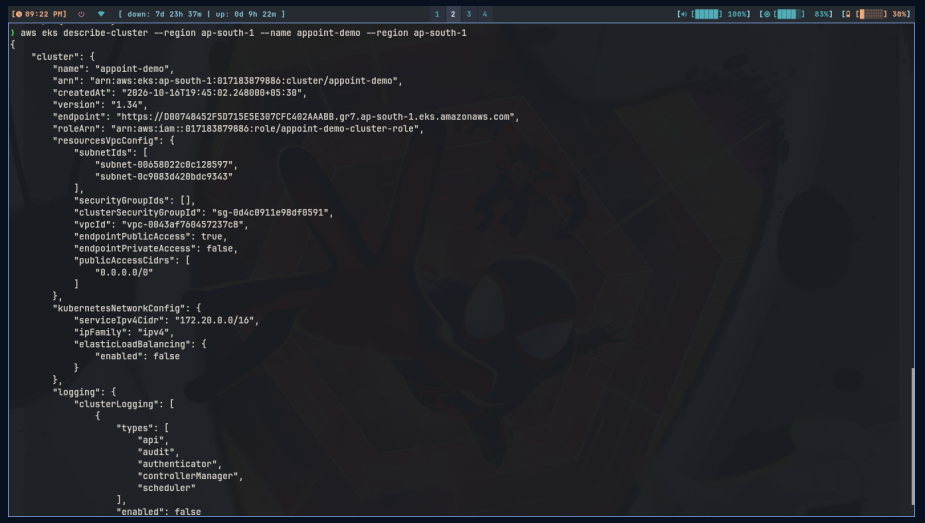
SERVICE MONITOR → PROMETHEUS TARGET UP → GRAFANA REQUEST RATE + P95 LATENCY

AWS INFRASTRUCTURE

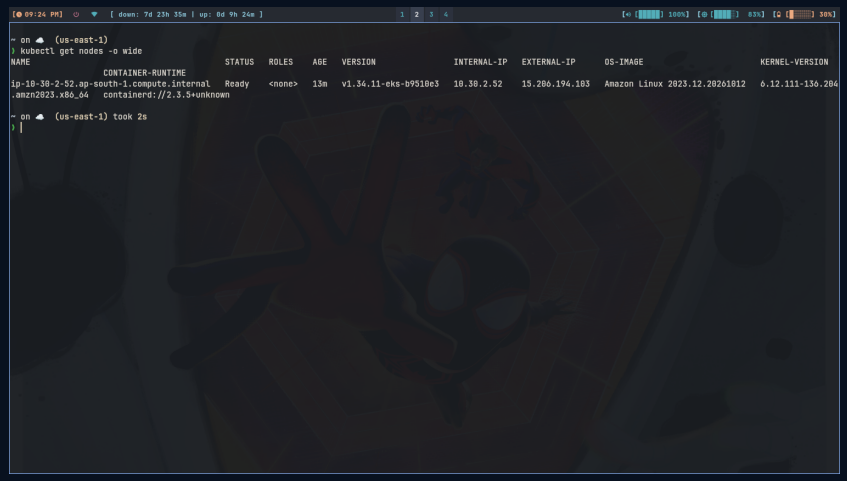
Terraform provisions a disposable, multi-AZ EKS foundation



```
❯ aws ec2 describe-vpcs --region ap-south-1 \
    --filters Name=tag:Name,Values=appoint-demo-vpc
{
    "Vpcs": [
        {
            "OwnerId": "017183879886",
            "InstanceTenancy": "default",
            "CidrBlockAssociationSet": [
                {
                    "AssociationId": "vpc-cidr-assoc-018ca7771e828a61e",
                    "CidrBlock": "10.30.0.0/16",
                    "CidrBlockState": {
                        "State": "associated"
                    }
                }
            ],
            "IsDefault": false,
            "Tags": [
                {
                    "Key": "ManagedBy",
                    "Value": "Terraform"
                },
                {
                    "Key": "Project",
                    "Value": "appoint"
                },
                {
                    "Key": "Name",
                    "Value": "appoint-demo-vpc"
                },
                {
                    "Key": "Environment",
                    "Value": "demo"
                }
            ],
            "BlockPublicAccessStates": {
                "InternetGatewayBlockMode": "off"
            },
            "VpcId": "vpc-0043af760457237c8",
            "State": "available",
            "CidrBlock": "10.30.0.0/16"
```

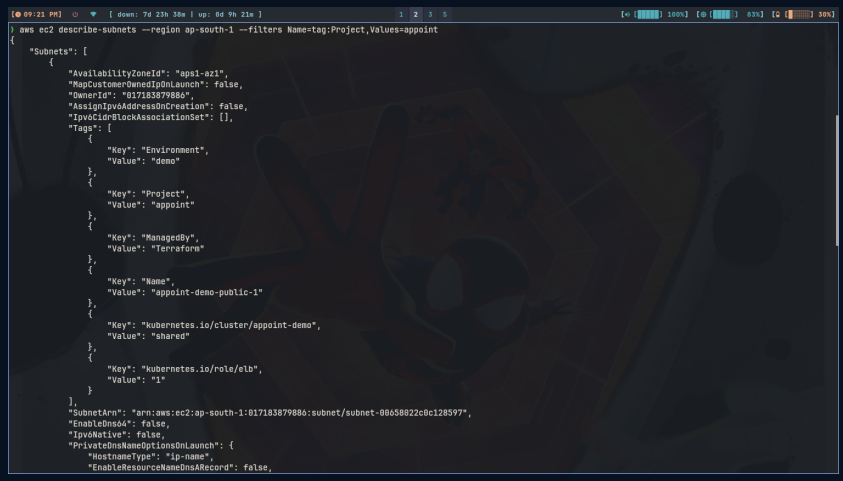


```
❯ aws eks describe-cluster --region ap-south-1 --name appoint-demo --region ap-south-1
{
    "cluster": {
        "name": "appoint-demo",
        "arn": "arn:aws:eks:ap-south-1:017183879886:cluster/appoint-demo",
        "createdAt": "2026-10-16T19:45:02.248000+05:30",
        "version": "1.34",
        "endpoint": "https://DB0748452F5D715E5E307CFC402AAABB.gr7.ap-south-1.eks.amazonaws.com",
        "roleArn": "arn:aws:iam::017183879886:role/appoint-demo-cluster-role",
        "resourcesVpcConfig": {
            "subnetIds": [
                "subnet-08658022c8c128597",
                "subnet-0c9883d428bd6c9343"
            ],
            "securityGroupIds": [],
            "clusterSecurityGroupId": "sg-0d4c0911e98df0591",
            "vpcId": "vpc-0043af760457237c8",
            "endpointPublicAccess": true,
            "endpointPrivateAccess": false,
            "publicAccessCidrs": [
                "0.0.0.0/0"
            ]
        },
        "kubernetesNetworkConfig": {
            "serviceIpv4Cidr": "172.20.0.0/16",
            "ipFamily": "ipv4",
            "elasticLoadBalancing": {
                "enabled": false
            }
        },
        "logging": {
            "clusterLogging": [
                {
                    "types": [
                        "api",
                        "audit",
                        "authenticator",
                        "controllerManager",
                        "scheduler"
                    ],
                    "enabled": false
```



```
~ on ☁  (us-east-1)
❯ kubectl get nodes -o wide
NAME                                        STATUS   ROLES    AGE   VERSION               INTERNAL-IP   EXTERNAL-IP      OS-IMAGE                         KERNEL-VERSION
                  CONTAINER-RUNTIME
ip-10-30-2-52.ap-south-1.compute.internal   Ready    <none>   13m   v1.34.11-eks-b9510e3   10.30.2.52    15.206.194.103   Amazon Linux 2023.12.20261012   6.12.111-136.204
.amzn2023.x86_64   containerd://2.3.5+unknown

~ on ☁  (us-east-1) took 2s
❯
```

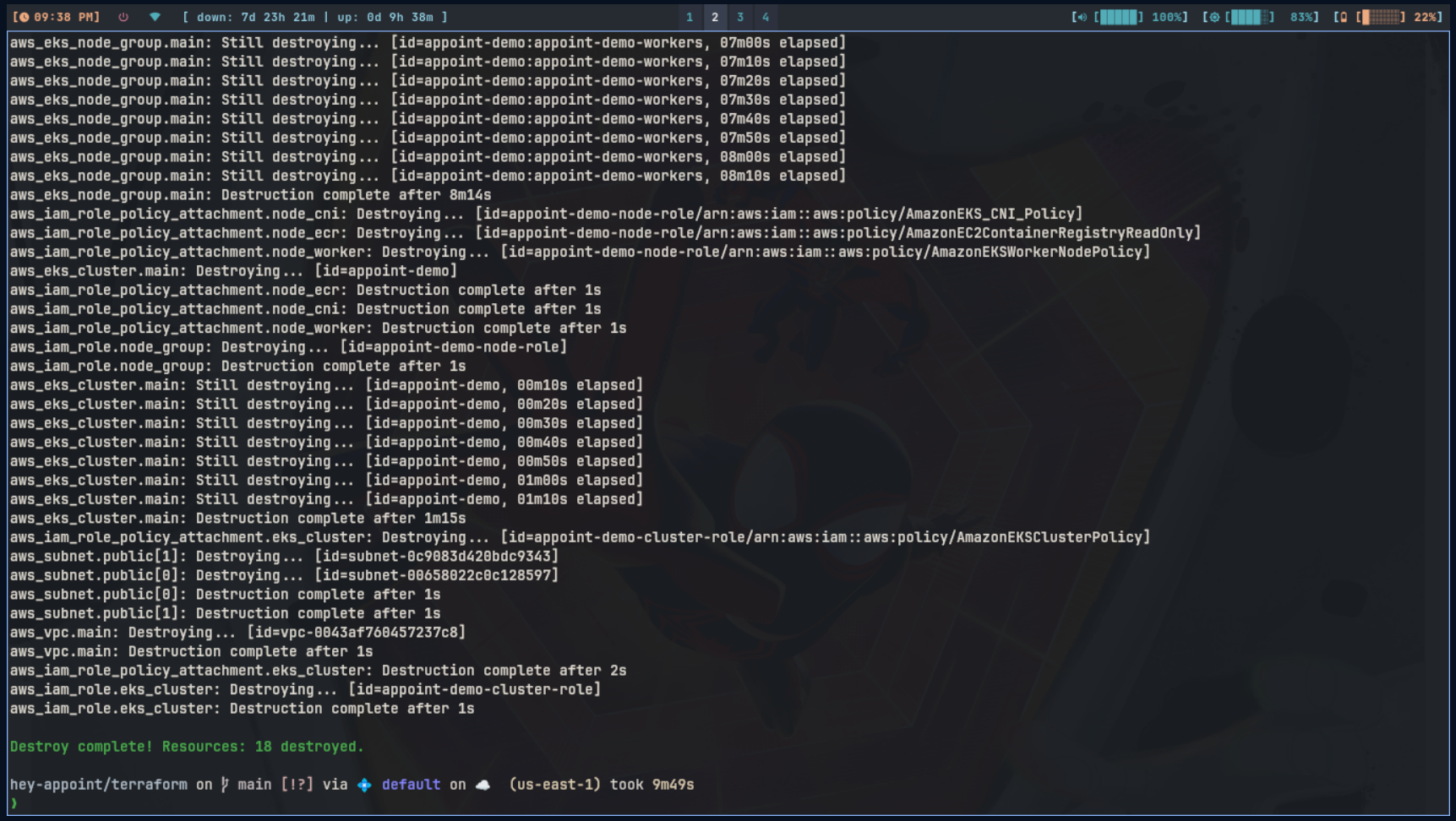


```
❯ aws ec2 describe-subnets --region ap-south-1 --filters Name=tag:Project,Values=appoint
{
    "Subnets": [
        {
            "AvailabilityZoneId": "aps1-az1",
            "MapCustomerOwnedIpOnLaunch": false,
            "OwnerId": "017183879886",
            "AssignIpv6AddressOnCreation": false,
            "Ipv6CidrBlockAssociationSet": [],
            "Tags": [
                {
                    "Key": "Environment",
                    "Value": "demo"
                },
                {
                    "Key": "Project",
                    "Value": "appoint"
                },
                {
                    "Key": "ManagedBy",
                    "Value": "Terraform"
                },
                {
                    "Key": "Name",
                    "Value": "appoint-demo-public-1"
                },
                {
                    "Key": "kubernetes.io/cluster/appoint-demo",
                    "Value": "shared"
                },
                {
                    "Key": "kubernetes.io/role/elb",
                    "Value": "1"
                }
            ],
            "SubnetArn": "arn:aws:ec2:ap-south-1:017183879886:subnet/subnet-08658022c8c128597",
            "EnableDns64": false,
            "Ipv6Native": false,
            "PrivateDnsNameOptionsOnLaunch": {
                "HostnameType": "ip-name",
                "EnableResourceNameDnsARecord": false,
```

VPC + two public subnets in separate AZs + EKS control plane + managed worker node group

EVIDENCE AND SAFE TEARDOWN

Every rubric claim is backed by a visible artifact



submission/ gallery • CI screenshots • AWS screenshots • Terraform destroy complete

KEY LEARNINGS AND NEXT STEPS

A platform that is reproducible, observable, and safe to tear down

- Automated quality gates make releases repeatable and auditable
- Infrastructure, delivery, and observability are documented together
- Non-root images, updates, and Trivy scanning reduce delivery risk
- Next evolution: ArgoCD GitOps sync, managed secrets, and TLS

READY TO DEMONSTRATE